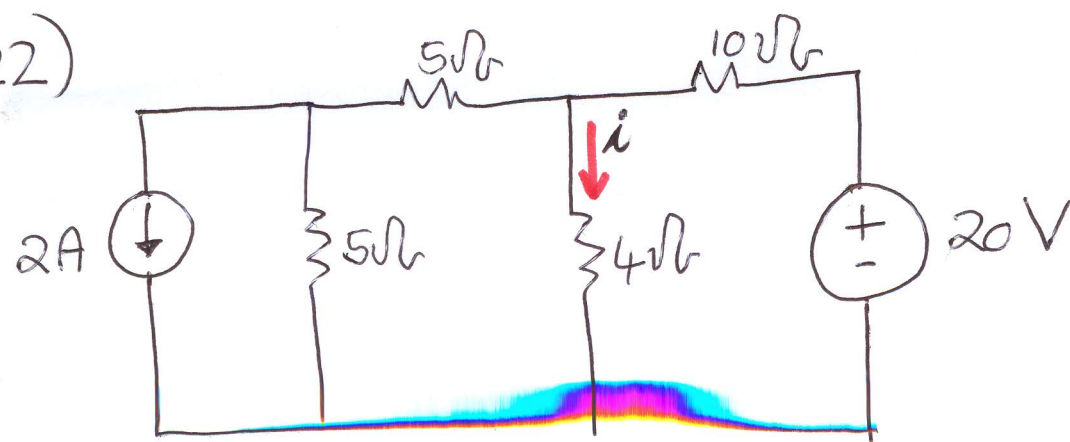


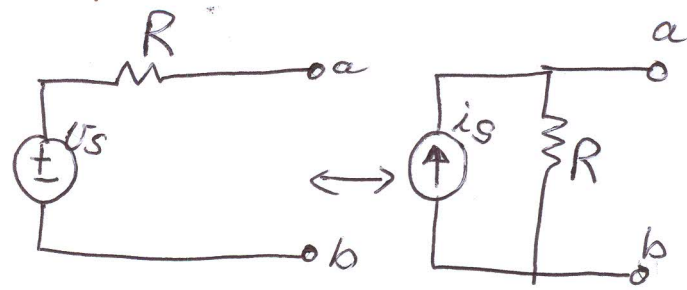
4.22)



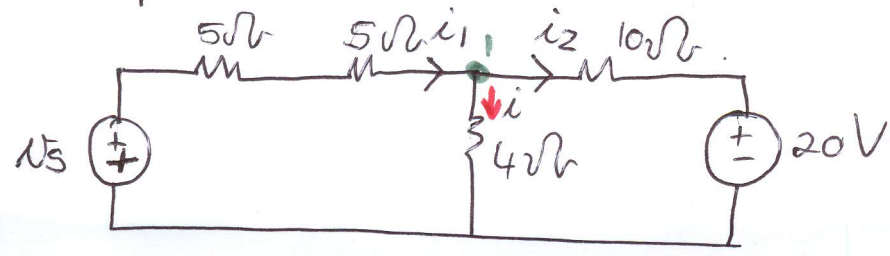
$$v_s = i_s R$$

use source transformation

find **i**?

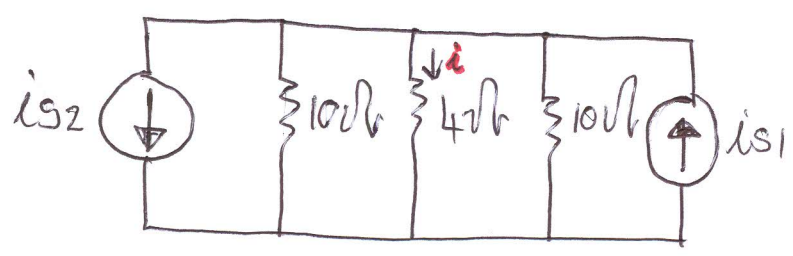


- a) Replace 2A $\rightarrow$ & $R=5\Omega$.



$$v_s = i_s R = (2)(5) = 10V$$

- b) Replace 20V $\rightarrow$ & 10Ω (i_{s1})
and $v_s = 10V \rightarrow$ & 10Ω ($5+5\Omega$ serie) - (i_{s2})



$$i_{s1} = \frac{20}{10} = 2A$$

$$i_{s2} = \frac{10}{10} = 1A$$

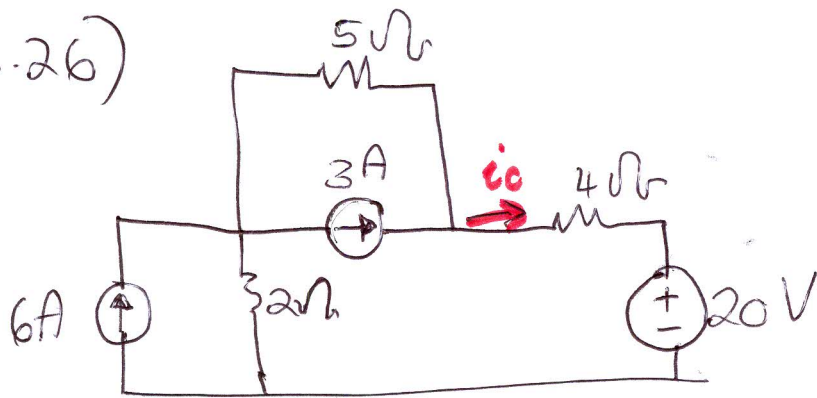
$$10\Omega // 10\Omega = 5\Omega$$

For i : Use current division

$$\therefore i_1 = \left(\frac{R_2}{R_1 + R_2} \right) i \quad (\text{equat. 2.43}) \quad \therefore \underline{i = 555.5 \text{ mA}} \rightarrow$$

where $\underline{i = i_{s1} - i_{s2} = 1A}$; $i_1 = i$; $R_2 = 5\Omega$; $R_1 = 4\Omega$

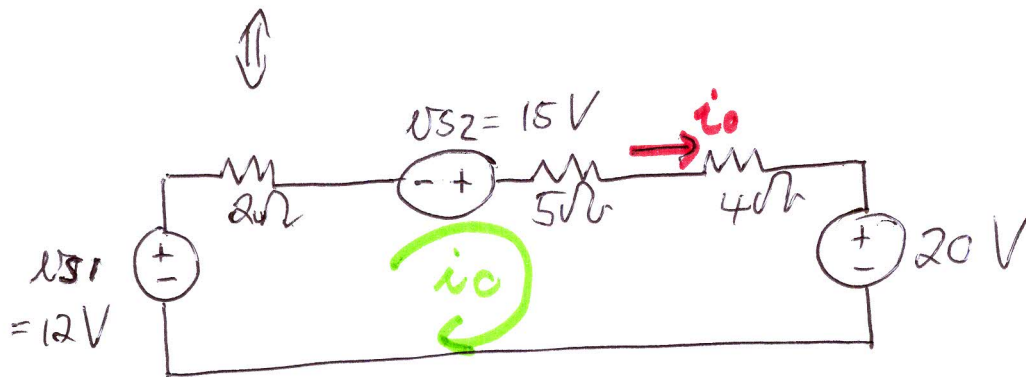
4.26)



② Use source transformation!

Find i_o ?

Transform both $\rightarrow$ to $\rightarrow$



$$V_{S1} = i_{S1} R = (6)(2) = 12V$$

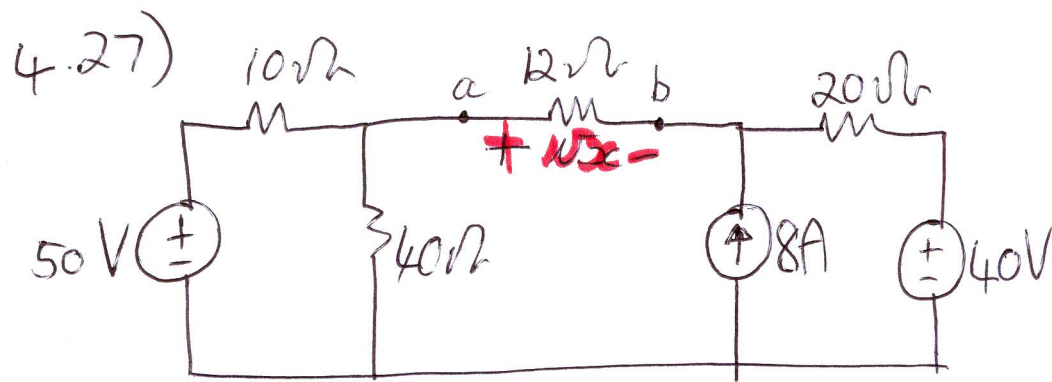
$$V_{S2} = i_{S2} R = (3)(5) = 15V$$

KVL : $-12 + 2i_o - 15 + 4i_o + 20 = 0$

$$11i_o = 7$$

$$\underline{i_o = 636.36 \text{ mA}}$$

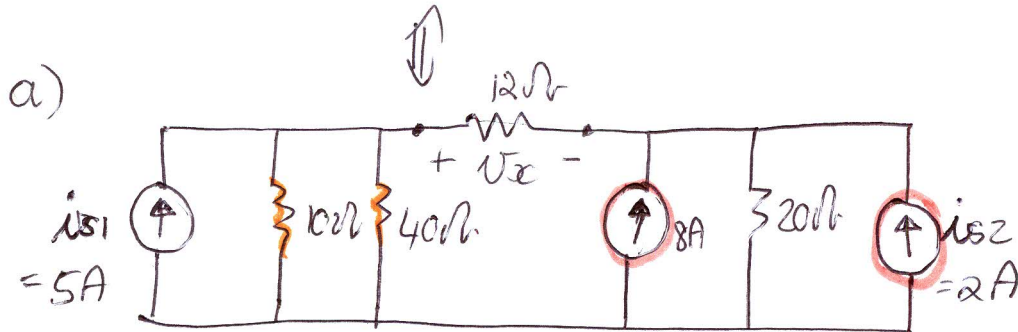
③



Apply Source transformation.

Find V_x ?

First transform both $\text{---} \text{---} \text{---}$ to $\text{---} \text{---} \text{---}$

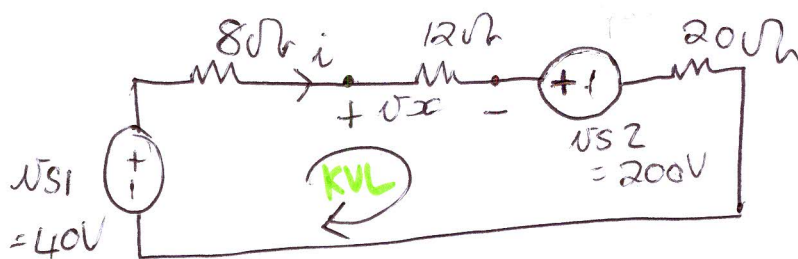


$$i_{s1} = \frac{V_s}{R} = \frac{50}{10} = 5A$$

$$i_{s2} = \frac{V_s}{R} = \frac{40}{20} = 2A$$

$$\underline{10\Omega // 40\Omega = 8\Omega} \quad \& \quad \underline{8A + 2A = 10A}$$

b) Now transform all $\text{---} \text{---} \text{---}$ to $\text{---} \text{---} \text{---}$



$$V_{s1} = i_{s1} R = (5)(8) = 40$$

$$V_{s2} = i_{s1} R = (10)(20) = 200$$

KVL: $V_x + 200 + 20i - 40 + 8i = 0 \dots \textcircled{1}$

at 12Ω $\text{---} \text{---} \text{---}$: $i = \frac{V_x}{12}$, replace in $\textcircled{1}$

$$V_x + 200 + 20\left(\frac{V_x}{12}\right) - 40 + 8\left(\frac{V_x}{12}\right) = 0$$

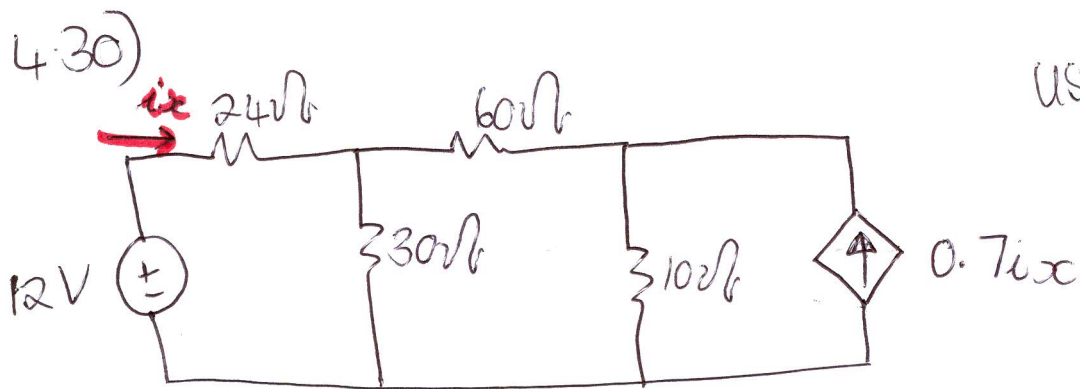
$$\frac{10}{3} V_x = -160$$

$$V_x = -48V$$

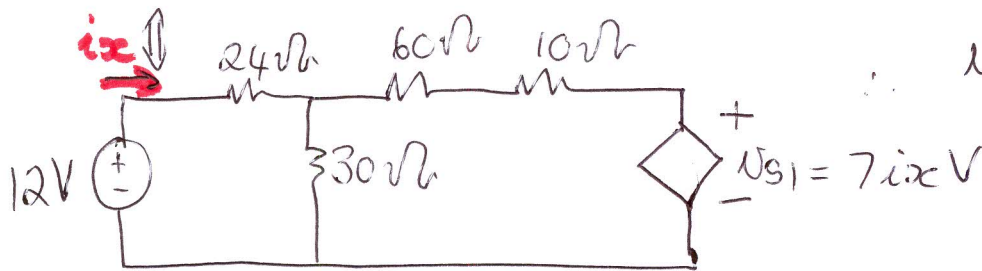


(4)

use source transformation

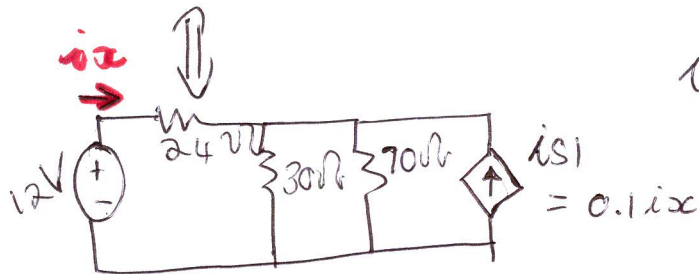


a) transform to



$$60 + 10 = 70 \Omega$$

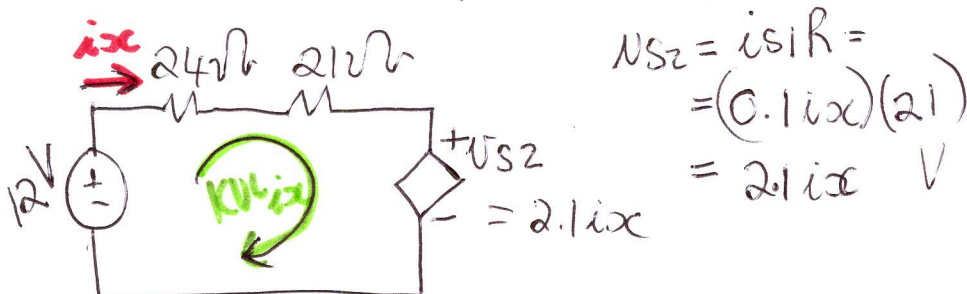
b) transform to



$$i_{s1} = \frac{V_{s1}}{R} = \frac{7i_x}{70} = 0.1i_x \text{ A}$$

$$30\Omega // 70\Omega = 21\Omega$$

c) transform to

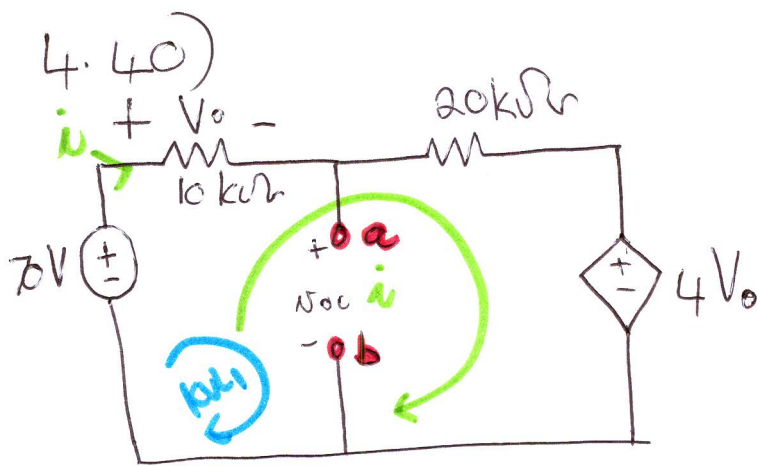


$$\text{KVL } i_x \equiv -12 + 24i_x + 21i_x + V_{s2} = 0$$

$$-12 + 45i_x + 2.1i_x = 0$$

$$i_x = \frac{12}{47.1} = 254.8 \text{ mA}$$

(5)



Find Thevenin equivalent
at a-b

$$V_{th} = V_{oc} \quad (\text{Voltage of o-c, linear circuit})$$

$$R_{th} = R_{in} \quad (\text{resistance of linear circuit, } \text{off})$$

For V_{th} ; normal KVL to obtain i :

$$-70 + (10k)i + (20k)i + 4V_O = 0$$

But $V_O = (10k)(i)$

$$\therefore -70 + (30k)i + (40k)i = 0$$

$$(70k)i = 70$$

$$\therefore i = \frac{70}{70000} = 1 \times 10^{-3} \text{ A}$$

$$V_{th} = V_{oc} \quad \text{KVL:}$$

$$\therefore -70 + (10000)i + V_{oc} = 0$$

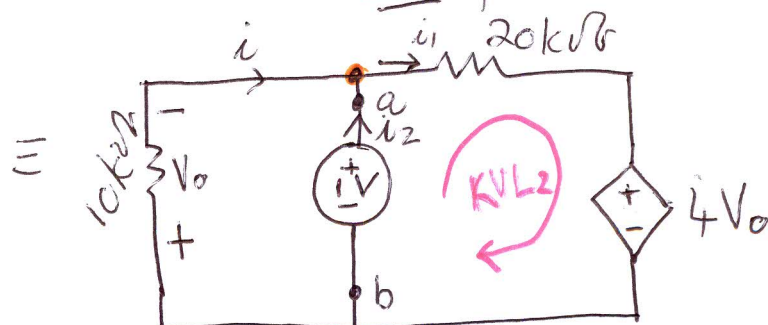
$$V_{oc} = 70 - 10$$

$$V_{oc} = 60 \text{ V} = V_{th}$$

b) For R_{th} All independent source off; thus 70V $\rightarrow$ short circuit

Apply (1V) source
at terminals

a b



(6)

from (b) circuit $V_o = -1V$

KVL₂: $-1 + (20k)i_1 + 4V_o = 0$

$$-1 + (20k)i_1 - 4 = 0$$

$$\therefore i_1 = 0.25 \text{ mA}$$

KCL $i + i_2 = i_1$

$$i_2 = i_1 - i$$

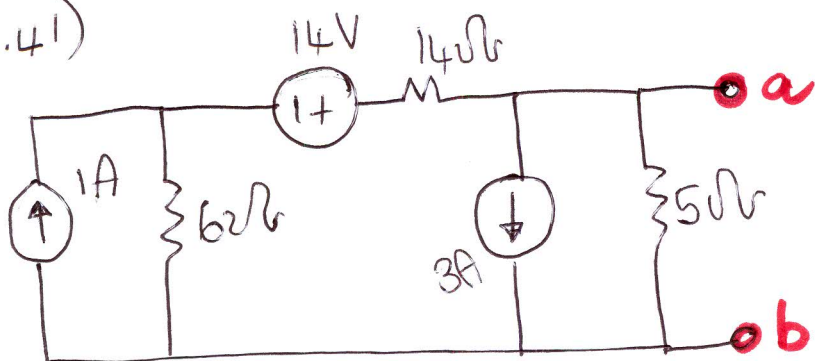
$$i_2 = (0.25 \text{ mA}) - \left(\frac{V_o}{10k} \right)$$

$$i_2 = 0.35 \text{ mA}$$

Thus **R_{th}** $= \frac{1(V)}{i_2} = 2.857 \text{ k}\Omega$

4.41)

⑦

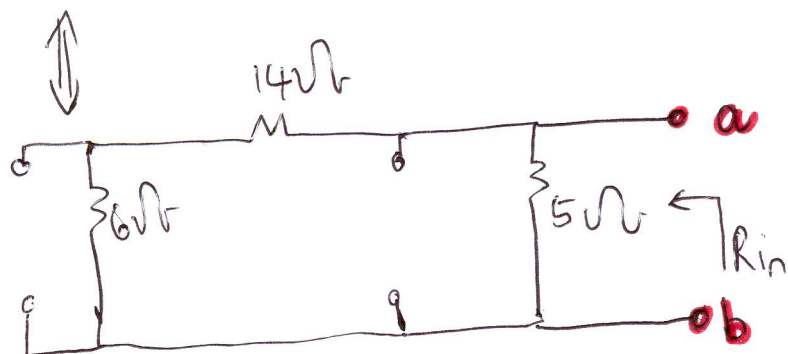


Find Thevenin
equivalent
of $a-b$

R_{th}: all independent sources off:

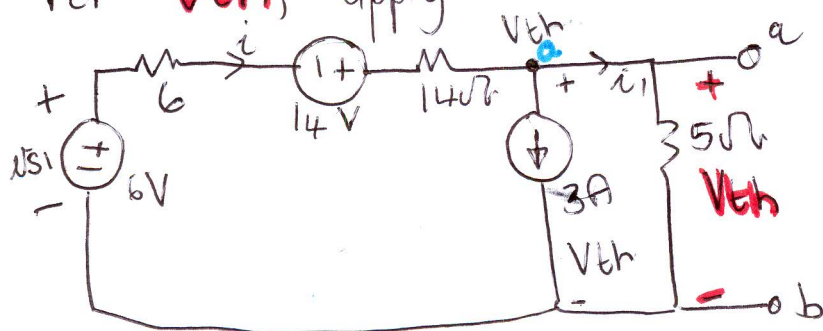
$\text{---} \bigcirc \text{---}$ $\equiv$ short circuit

$\text{---} \bigcirc \text{---}$ $\equiv$ open circuit



$$R_{th} = R_{in}$$
$$= (14 + 6) \parallel 5$$
$$= \frac{(20)(5)}{25} = 4 \Omega$$

For V_{th} , apply source transformation



$$V_{S1} = (i_S)R = 6V$$

KCL_{node}: $i = 3 + i_1$

$$\frac{(6+14) - V_{th}}{14 + 6} = 3 + \frac{V_{th}}{5}$$

$$5(6+14 - V_{th}) = 300 + 20 V_{th}$$

$$V_{th} = -8 \text{ V}$$

4.47)

The circuit diagram shows a 50V DC voltage source in series with a 12Ω resistor. This is followed by a 60Ω resistor in parallel with a dependent voltage source of $2V_x$. The output terminals are labeled a and b . The voltage across the 60Ω resistor is V_x , and the voltage across the dependent source is $2V_x$. The output voltage is V_{ab} . A pink oval highlights the 12Ω resistor and the dependent source.

8

For V_{th} : $V_{th} = V_{ab} = V_{ac}$ (parallel branches)

KCL_a:

$$\dot{I} = \dot{I}_Z + 2V_{OC}$$

$$\frac{+50 - V_{ox}}{12} = \frac{V_{ox}}{60} + 2V_{ox}$$

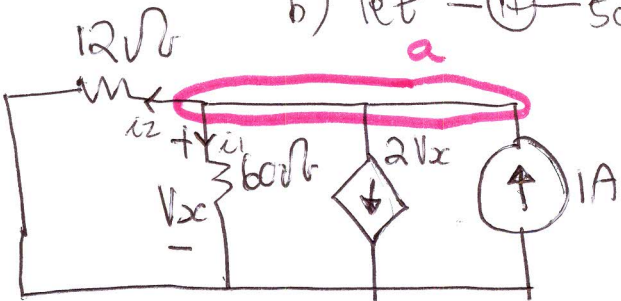
$$\therefore 3000 - 60V_x = 12V_x + 1440 \text{ Vsc}$$

$$1512Vx = 3000$$

$$V_x = V_{ab} - \underline{V_{th}} = 1.9841 \text{ V}$$

For Rth; a) add/insert  test source between a and b

b) let 14Ω short circuit



KCL_a: $1 = 2V_x + i_1 + i_2$

$$1 = 2Vx + \frac{Vx}{12} + \frac{Vx}{60}$$

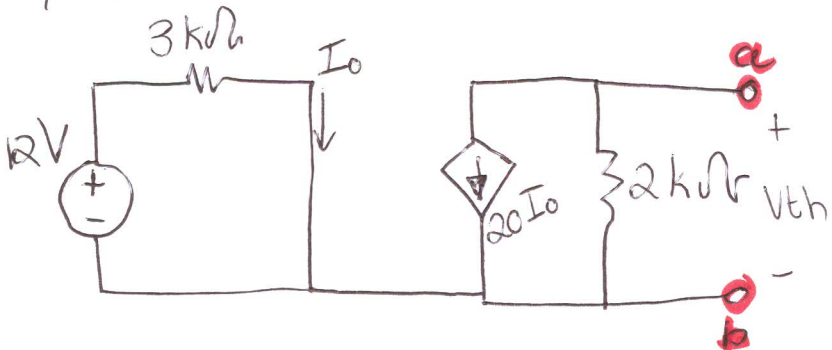
$$720 = 1440 V_x + 60 V_x + 12 V_x$$

$$\therefore V_{DC} = 0.4762 \text{ V}$$

Thus $R_{th} = \frac{V_{oc}}{I} = 0.4762 \Omega$

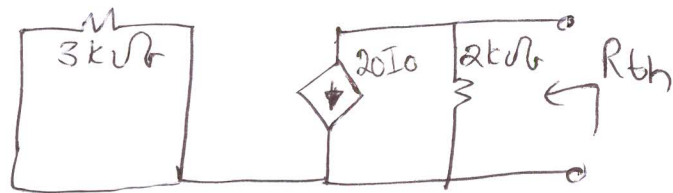
4.52)

(9)



Obtain thevenin equivalent at a-b

For R_{th} : $\overset{12V}{\text{---} \text{---} \text{---}}$ is short circuit



$$\therefore I_o = 0$$

$$\therefore R_{th} = 2k\Omega$$

For V_{th}

$$I_o = \frac{12}{3k} = 4 \text{ mA}$$

$$\therefore V_{th} = (-20I_o)(2k) = -160 \text{ V}$$